

# $L$ -functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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## Abstract

We give a single manifestly finite, regulator-free functional realizing in one object  $L$ -functions quasi-periods and period polynomials of meromorphic modular forms on  $\mathrm{SL}_2(\mathbb{Z})$ , with generic poles throughout the upper half-plane  $\mathbb{H}$  and at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ . Additionally, our construction preserves holomorphicity. It is a two-segment contour integral,  $L^*(f, s) = L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S)$ , where  $L_T^*$  and  $L_S^*$  have distinct integration kernels, and  $\gamma_{\tau_0}^S$  is a path in  $\mathbb{H}$  from  $S(\tau_0)$  to  $\tau_0$ , and the same for  $\gamma_{\tau_0}^T$ . For weakly holomorphic forms our  $L$ -function is independent of  $\tau_0$  and of the path, and reproduces classical  $L$ -functions for holomorphic modular forms, reproduces the  $L$ -functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined, and reproduces Brown's quasi-periods [2]. For properly meromorphic modular forms with poles inside the interior of  $\mathbb{H}$ , our  $L$ -function depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our  $L$ -functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [3].

## 1 Introduction and preliminary concepts

Our primary goal in this note is to put forth a unifying generalization of the  $L$ -function associated to a given modular form. Our  $L$ -function naturally encompasses the classical case of  $L$ -functions associated to holomorphic modular forms [4], and seamlessly unifies Ref. [1]'s extension to  $L$ -functions of modular forms with poles at the cusp with Ref. [3]'s extension to  $L$ -functions of modular forms with poles at finite locations, exclusively away from cusps. Secondly, we show that our  $L$ -functions reproduce the numerical results for quasi-periods associated to weak cusp forms [2].

*Note:* Throughout this document, we refer to  $L$ -integrals and  $L$ -functions. The classical  $L$ -function-as-Dirichlet-series which has been suitably extended to the complex  $s$  plane is simply not defined when the  $L$ -function in question has poles either at the cusp or at finite locations within the upper half-plane. So, if there is ever a doubt, we state clearly: *any  $L$ -function in this note is defined by an  $L$ -integral*. The focus of this note is to define  $L$ -integrals for any integer weight  $\mathrm{SL}_2(\mathbb{Z})$  modular form which may have poles at cusps or away from cusps.

### 1.1 Modular forms

Write  $\mathrm{Re}(z)$  and  $\mathrm{Im}(z)$  for the real and imaginary parts of a complex number  $z \in \mathbb{C}$ . Let  $\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$  denote the upper half-plane.  $\mathrm{SL}_2(\mathbb{Z})$  is the group generated by  $T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ .

A *holomorphic modular form* of weight  $k$  for  $\mathrm{SL}_2(\mathbb{Z})$  is a holomorphic function  $f : \mathbb{H} \rightarrow \mathbb{C}$  such that  $f(\tau+1) = f(-\frac{1}{\tau})\tau^{-k} = f(\tau)$  for all  $\tau \in \mathbb{H}$ , and  $f(\tau) = \mathcal{O}(1)$  as  $\mathrm{Im}(\tau) \rightarrow \infty$ , and which has no poles at finite locations within  $\mathbb{H}$ . By virtue of the periodicity  $f(\tau) = f(\tau+1)$  and holomorphicity,

each holomorphic modular form has a q-series decomposition,

$$f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau} = \sum_n c_f(n) q^n \quad , \quad q := e^{2\pi i \tau} \quad . \quad (1)$$

In order for  $f$  to be bounded as  $\tau \rightarrow i\infty$ , we conclude that  $c_f(n) = 0$  for all  $n < 0$ , i.e. that  $f = \sum_{n \geq 0} c_f(n) q^n$ ; for such modular forms  $f$  we write  $f \in M_k$ . If a modular form  $f$  additionally has a vanishing constant term, i.e. if  $c_f(0) = 0$ , then  $f$  is a *cusp form*, and we write  $f \in S_k$ . Clearly,  $S_k \subset M_k$ . Further,  $M_k$  is a finite-dimensional vector-space, as is  $S_k$ . Holomorphic  $\mathrm{SL}_2(\mathbb{Z})$  modular forms must have even integer weights, and  $k \geq 4$  [4].

A product of modular forms of weights  $k_1$  and  $k_2$  is a modular form of weight  $k_1 + k_2$ , so  $M_* := \bigoplus_k M_k$  is naturally a (graded) ring. This graded ring is freely generated by the two modular forms  $E_4 \in M_4$  and  $E_6 \in M_6$ , the  $k = 4, 6$  specializations of the *weight  $k$  Eisenstein series*  $E_k$ , which may be defined as follows:

$$E_k := \frac{1}{2} \sum_{(m,n)=1} \frac{1}{(m+n\tau)^k} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \quad , \quad (2)$$

where  $\sum_{(m,n)=1}$  denotes summing over all relatively prime  $(m, n) \in \mathbb{Z}^2$ , the Bernoulli number  $B_n$  is the coefficient of  $t^n/n!$  of the Laurent expansion of  $t/(e^t - 1)$  about  $t = 0$ , and  $\sigma_m(n) := \sum_{d|n} d^m$  is the Divisor function where  $d|n$  means  $n/d \in \mathbb{N}$ . In short,  $M_* = \mathbb{C}[E_4, E_6]$ . We now define the *modular discriminant*:

$$\Delta := \frac{E_4^3 - E_6^2}{1728} = q + O(q^2) \in S_{12} \quad . \quad (3)$$

Holomorphicity is most canonically relaxed via allowing a modular form to have a pole at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ , i.e. for a weight  $k$  modular form for  $\mathrm{SL}_2(\mathbb{Z})$  to have negative powers of  $q$  in its q-series expansion. In this case,  $f$  diverges as  $\tau \rightarrow i\infty$  as  $f \sim e^{-2\pi N \tau}$ , where  $-N$  is the largest negative number such that  $c_f(n)$  is non-zero. Such a modular form  $f$  is called a *weak modular form*, and we write  $f \in M_k^!$ . If a weak modular form  $f$  has a vanishing coefficient of  $q^0$ , i.e. if  $c_f(0) = 0$ , it is called a *weak cusp form* and we write  $f \in S_k^!$ . Clearly,  $S_k^! \subset M_k^!$ ,  $S_k \subset S_k^!$  and  $M_k \subset M_k^!$ . Two important weak cusp forms in our paper are as follows:

$$J := \frac{E_4^3}{\Delta} - 744 = \frac{1}{q} + O(q) \in S_0^! \quad , \quad (4)$$

$$\widehat{\Delta} = (J^2 + a_1 J + a_0) \Delta := \frac{1}{q} + O(q^2) \in S_{12}^! \quad , \quad (5)$$

where  $a_0, a_1$  are integers which exist to ensure the q-series gap that defines  $\widehat{\Delta}$  [5].

In this work, we further relax from even weakly holomorphic modular forms, and consider *meromorphic modular forms* of weight  $k$  which are quotients:  $f = \frac{f_1}{f_2}$  where  $f_i \in M_{k_i}$ , the denominator  $f_2$  is not identically zero, and  $k = k_1 - k_2$ . We write  $F_k$  for the vector space of meromorphic modular forms of weight  $k$ , and set  $F_* := \bigoplus_k F_k$ . Then  $F_k$  vanishes unless  $k$  is an even integer, and  $F_*$  may be regarded as a “graded ring of fractions” of  $M_*$ .

## 1.2 L-functions for holomorphic modular forms and periods for cusp forms

For any  $f \in S_k$ , one may define the associated L-function:

$$L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s) \stackrel{f \in S_k}{:=} \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty f(it) t^{s-1} dt = \sum_{n \geq 1} \frac{c_f(n)}{n^s} \quad . \quad (6)$$

A similar definition applies for  $f \in M_k$ . (The classical definition of  $L^*(f, s)$  for  $f \in M_k$  is the focus of Remark 3.6, below.) Similarly, one may define a period polynomial for  $f \in S_k$  in terms of this L-function,

$$r_f(X, Y) := (2\pi i)^{k-1} \sum_{\ell=0}^{k-2} (-1)^\ell \binom{k-2}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{k-2-\ell} Y^\ell. \quad (7)$$

This polynomial,  $r_f(X, Y) \in V_{k-2} := \mathbb{C}[X, Y]_{k-2}$ , has a natural action under  $\mathrm{SL}_2(\mathbb{Z})$ ,  $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$  for  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ . This period polynomial can be decomposed into a basis of highly constrained polynomials, which are annihilated by  $|(1+T)$  and  $|(1+U+U^2)[6]$ , where  $U := TS$  generates an order-3 cyclic subgroup of  $\mathrm{SL}_2(\mathbb{Z})$ . We define the *periods* of a cusp form to be the coefficients that localize  $r_f$  within this highly constrained sub-space of  $V_{k-2}$ .

**Why this paper?** As classically computed, these L-functions and associated period polynomials (when defined in terms of the special values of the associated L-function) are not defined when  $f$  has poles either at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$  or within the interior of  $\mathbb{H}$ . As stated above, there are extant, but partial and non-overlapping, extensions of the definition of  $L(f, s)$  to  $f \in M_k^!$  [1] or  $f \in F_k$  when  $f$  is regular at the cusp of  $\mathrm{SL}_2(\mathbb{Z})$  [3]. In this note, we provide a construction that unifies these disparate extensions. We start with generic L-integrals in section 2, then specialize to  $f \in M_k^!$  in section 3, and finally explore the more general context of  $f \in F_k$  in section 4.

## 2 A finite L-integral

**Definition 2.1.** Let  $\tau_0 \in \mathbb{H}$  and  $f \in F_k$ , let  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ,  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  denote the generators of  $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ , and let  $f^{-1}(\infty)$  be the locus of all poles of  $f \in \mathbb{H}$  (inclusive of the cusp of  $\mathrm{SL}_2(\mathbb{Z})$ ). The  $S$ - and  $T$ -segments at  $\tau_0$  are smooth paths in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$  from  $\sigma^{-1}\tau_0$  to  $\tau_0$ ,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0. \quad (8)$$

Avoiding poles is automatic for  $f \in M_k^!$ , as long as the base-point  $\tau_0$ , is not at a cusp of  $\mathrm{SL}_2(\mathbb{Z})$ , i.e. there is only the trivial homotopy class. When  $f \in F_k$ , and  $f$  has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

**Definition 2.2.** Let  $\zeta(s, x) = \sum_{m \geq 0} (m+x)^{-s}$  as a function  $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \mathbb{N})_x \rightarrow \mathbb{C}$  denote the Hurwitz zeta function (meromorphic in  $s$ , holomorphic in  $x \in \mathbb{H}$ ). We define Hurwitz  $T$ -kernel as

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1). \quad (9)$$

**Definition 2.3** (Generic L-integral). Let  $f \in F_k$ ,  $\tau_0 \in \mathbb{H}$ , and  $s \in \mathbb{C}$ . Further, let  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$  be the  $S$ - and  $T$ -segments at  $\tau_0$ , as in Definition 2.1, and let  $k_T$  be as defined in Definition 2.2. The L-integral of  $f$  is

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left( \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right) \quad (10)$$

We use this  $L$ -integral to define a generalized  $L$ -function:

**Definition 2.4** (Generic L-function). Let  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  be as defined in Definition 2.3. We define the generalized  $L$ -function of  $f \in F_k$  to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (11)$$

Having defined the L-integral and L-function, we prove they are manifestly finite for  $f \in F_k$ :

**Lemma 2.5** (Finiteness of L-integrals). *For  $f \in F_k$ , both integrals in Definition 2.3 are finite for every  $s \in \mathbb{C}$ , every  $\tau_0 \in \mathbb{H}$ , and every choice of paths  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ .*

*Proof.* Both segments are compact and lie in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , so  $f$  is continuous on them;  $\tau^{s-1}$  is continuous on  $\mathbb{H}$ , and  $\tilde{k}_T(\tau, s)$  is continuous there since  $\zeta(\nu, \tau + 1)$  is holomorphic in  $\tau \in \mathbb{H}$  (Definition 2.2). A continuous integrand on a compact path is integrable, for every  $s \in \mathbb{C}$ .  $\square$

We now turn to path-dependence and  $\tau_0$ -independence of the L-integral and the L-function of  $f \in F_k$ . This holds for all  $f \in M_k^!$ :

**Proposition 2.6** (Basepoint & path independence). *For  $f \in M_k^!$  the value  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\tau_0 \in \mathbb{H}$  and of the paths: with no interior poles,  $\mathbb{H}$  is simply connected, so all admissible contours are homotopic. We write  $L^*(f, s)$  for the common value and  $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$ .*

*Proof.* With no interior poles the integrands  $f(\tau)\tau^{s-1}$  and  $f(\tau)\tilde{k}_T(\tau, s)$  are holomorphic on  $\mathbb{H}$ , which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of  $\tau_0$  show that variations in  $\tau_0$  for the S-term and T-term in  $L^*(f, s)$  cancel, i.e. that  $\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = -\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T)$ . For  $L_T^*$ , note that the  $T$ -segment runs from  $T^{-1}\tau_0 = \tau_0 - 1$  to  $\tau_0$  — two endpoints a unit apart, moving together with  $\tau_0$  — so by Leibniz's rule and the  $T$ -periodicity  $f(\tau_0 - 1) = f(\tau_0)$ ,

$$\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (12)$$

where  $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$ . This is the whole role of  $\delta_T$ : the unit gap between the endpoints turns the  $\partial_{\tau_0}$  boundary term into a unit shift of the kernel. That shift is elementary — the kernel was built for it — since  $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$  gives, on both terms at  $x = \tau + 1$ ,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s} \quad (13)$$

$$\implies \partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (14)$$

The  $S$ -segment runs  $-1/\tau_0 \rightarrow \tau_0$ ; its lower endpoint moves with  $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$ , so by Leibniz, the modularity  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ , and the phase  $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)}\tau_0^{1-s}$ ,

$$\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = f(\tau_0)\tau_0^{s-1} - f(-1/\tau_0)(-1/\tau_0)^{s-1}\tau_0^{-2} = f(\tau_0)[\tau_0^{s-1} - e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (15)$$

Thus  $\tau_0$ -variations are equal and opposite,  $\partial_{\tau_0} L_S^* = -\partial_{\tau_0} L_T^*$ . Thus  $\partial_{\tau_0} L^* = \partial_{\tau_0} L_S^* + \partial_{\tau_0} L_T^* = 0$ .  $\square$

**Remark 2.7** (Recovering the classical L-integral for  $f \in S_k$ ). *Let  $f \in S_k$ . As  $\tau_0 \rightarrow i\infty$ , the  $S$ -segment maps onto the standard L-function contour,  $[0, i\infty]$ . Simultaneously, as  $f(it) \sim e^{-2\pi t}$  as  $t \rightarrow \infty$ , the  $T$ -segment vanishes exponentially. Thus, in this limit,  $L^*(f, s)$  exactly maps onto the classical L-function integral [4]:*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (16)$$

### 3 $L$ -functions for weak modular forms

**Proposition 3.1.** Let  $\Gamma(s, x) := \int_x^\infty e^{-t} t^{s-1} dt$  denote the incomplete Gamma function. For  $n \neq 0$ ,  $\nu \in \mathbb{C}$ , and  $z \in \mathbb{H}$ ,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (17)$$

*Proof.* Write  $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$  and shift  $u \mapsto u+m$  in the  $m$ -th term; as  $e^{2\pi i n(u+m)} = e^{2\pi i n u}$  for integer  $m$ , the unit segments stack into one horizontal ray,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{z+m}^{z+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (18)$$

Substitute  $w = -2\pi i n v$ , so that  $v = iw/(2\pi n)$ ,  $dv = i dw/(2\pi n)$ , and  $e^{2\pi i n v} = e^{-w}$ : the lower endpoint becomes  $w = -2\pi i n z$ , and for  $n > 0$  the ray rotates to  $w \rightarrow +\infty$  along the positive reals — legal, since  $e^{-w} w^{-\nu}$  decays in the right half-plane, so the closing arc vanishes. With  $v^{-\nu} = i^{-\nu} (2\pi n)^{\nu} w^{-\nu}$ ,

$$\int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} (2\pi n)^{\nu-1} \int_{-2\pi i n z}^{\infty} e^{-w} w^{-\nu} dw = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (19)$$

For  $n < 0$  the ray instead grows as  $\text{Im} v \rightarrow \infty$  and the rotation is illegal; but both sides are analytic in  $n$  and agree for  $n > 0$ , so the identity continues to  $n < 0$  with  $\Gamma$  on its principal branch. At  $n = 0$  the prefactor  $(2\pi n)^{\nu-1}$  is singular; that case is handled separately.  $\square$

**Theorem 3.2** (The  $L$ -function for  $f \in M_k^1$ ). For  $f \in M_k^1$ , we may set  $\tau_0 = i$ , and prove

$$L^*(f, s) = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (20)$$

Here, trivially,  $L^*(f, s) = i^k L^*(f, k-s)$ . The  $L$ -function is  $L(f, s) := (2\pi)^s L^*(f, s)/\Gamma(s)$ .

*Proof.* By path independence and base-point independence from Proposition 2.6, we may freely set  $\tau_0 = i$ . Here,  $L_S^*(f, s; \gamma_{\tau_0}^S) = 0$  (the  $S$ -contour vanishes in length and there are no nontrivial homotopy classes as  $f \in M_k^1$  has no poles which a finite contour can wrap), and the  $L$ -integral simplifies dramatically:

$$L^*(f, s) = L_T^*(f, s; \gamma_i^T) = e^{-\pi i s/2} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (21)$$

We then expand  $f$  in terms of its  $q$ -series,  $\sum_n c_f(n) e^{2\pi i n \tau}$ , and substitute the  $u$ -variable of Proposition 3.1 as  $u = \tau + 1$ , and note that  $e^{2\pi i n(u-1)} = e^{2\pi i n u}$ . When  $n \neq 0$ , straightforward substitutions of  $\nu, z$  into Proposition 3.1 suffice:  $1-\nu = s$  gives the contribution of the first term in  $\tilde{k}_T(\tau, s)$ , and  $1-\nu = k-s$  gives the second:

$$e^{-\pi i s/2} \left( e^{\pi i s/2} \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} - e^{i\pi(s-1)} e^{\pi i(k-s)/2} \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (22)$$

We may then evaluate the  $n = 0$  mode's contribution follows from noting the closed form solution for the definite integral,

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma-1}. \quad (23)$$

This result follows from noting  $\partial_a(\zeta(\sigma-1, a)/(1-\sigma)) = \zeta(\sigma, a)$  and  $\zeta(\sigma-1, a+1) - \zeta(\sigma-1, a) = -a^{1-\sigma}$ . Applying this to the two distinct terms of  $\tilde{k}_T(\tau, s)$  at  $\sigma = 1-s$  and  $\sigma = 1-(k-s)$  gives

$$e^{-\pi is/2} \int_{i-1}^i c_f(0) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right). \quad (24)$$

Combining the  $n=0$  contributions with the  $n \neq 0$  contributions reproduces Eq. (20). Finally, the functional equation,  $L^*(f, s) = i^k L^*(f, k-s)$ , follows by direct inspection.  $\square$

### 3.1 Explicit S- and T-contours for general $\tau_0$

**Lemma 3.3** ( $L_T^*$  for  $M_k^!$ ). *For  $f \in M_k^!$ , and  $\tau_0 \in \mathbb{H}$  away from cusps,  $L_T^*(f, s; \gamma_{\tau_0}^T)$  evaluates to:*

$$L_T^*(f, s; \gamma_{\tau_0}^T) = -c_f(0) \left( \frac{(\tau_0/i)^s}{s} + i^k \frac{(\tau_0/i)^{k-s}}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (25)$$

*Proof.* Expand  $f = \sum_n c_f(n) q^n$  on the compact segment  $[\tau_0-1, \tau_0]$ , and restrict attention to the sub-sum,  $\sum_{n \neq 0}$ . For each such  $n \neq 0$ , apply the Proposition 3.1 in  $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$  at  $\nu = 1-s$  and  $\nu = s-(k-1)$ . The phases for the second term work-out identically as in the proof of Theorem 3.2.

For  $n=0$ , the constant  $c_f(0)$  multiplies  $e^{-\pi is/2} \int_{\tau_0-1}^{\tau_0} \tilde{k}_T(\tau, s) d\tau$ ; the Hurwitz antiderivative  $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1) -$  from  $\partial_\tau \zeta(\sigma-1, \tau+1) = -(\sigma-1)\zeta(\sigma, \tau+1)$  and the shift  $\zeta(\sigma-1, \tau_0+1) - \zeta(\sigma-1, \tau_0) = -\tau_0^{1-\sigma}$  — applied to the two halves of  $\tilde{k}_T$  at  $\sigma = 1-s$  and  $\sigma = s-k+1$  yields  $-e^{-\pi is/2} \tau_0^s/s$  and  $-e^{\pi is/2} \tau_0^{k-s}/(k-s)$  respectively.  $\square$

**Corollary 3.4** ( $L_T^*$  as  $\tau_0 \rightarrow i\infty$ ). *Let  $f \in S_k^!$ , where  $f = \sum_{n \geq -N} c_f(n) q^n$  for some  $-N \in \mathbb{Z}$ . The T-contour's contribution to  $L^*(f, s)$  is given by Lemma 3.3. As  $\tau_0 = it$  goes to the cusp  $i\infty$ , the T-contour scales exactly as  $f(it)$  scales:  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(it) \sim e^{2\pi Nt}$ .*

*Proof.* For the asymptotics,  $\Gamma(\sigma, z) \sim z^{\sigma-1} e^{-z}$  as  $|z| \rightarrow \infty$  for  $|\arg z| < \frac{3\pi}{2}$  [10, §8.11] gives  $|c_f(n)|$ -weighted go as  $e^{-2\pi n \text{Im}(\tau_0)}$ . The smallest mode  $n$  in the q-series, i.e.  $n = -N$ , dominates the scaling of  $f(it)$  and of  $L_T^*(f, s; \gamma_{it}^T)$ . Combining these results yields the claimed asymptotics.  $\square$

Now, exploiting  $\tau_0$ -independence of  $L^*(f, s)$ , its explicit value as given in Theorem 3.2, and the explicit T-contribution  $L_T^*(f, s; \gamma_{\tau_0}^T)$  trivially implies clean, closed-form expressions for  $L_S^*(f, s; \gamma_{\tau_0}^S)$ :

**Corollary 3.5** ( $L_S^*$  for  $M_k^!$ ). *For  $f \in M_k^!$  and  $\tau_0 \in \mathbb{H}$ ,  $L_S^*(f, s; \gamma_{\tau_0}^S)$  evaluates to*

$$L_S^*(f, s; \gamma_{\tau_0}^S) = c_f(0) \left( \frac{(\tau_0/i)^s - 1}{s} + i^k \frac{(\tau_0/i)^{k-s} - 1}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left( \frac{\Gamma(s, 2\pi n) - \Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n) - \Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (26)$$

*Proof.* Recall  $L^*(f, s) - L_T^*(f, s; \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S)$ . Now,  $L^*(f, s)$  for  $f \in M_k^!$  is given by Theorem 3.2, and  $L_T^*(f, s; \gamma_{\tau_0}^T)$  is given by Lemma 3.3. Their difference yields Eq. (26).  $\square$

### 3.2 Classical Mellin recovery and canonical uniqueness.

As discussed in Remark 2.7, for  $f \in S_k$ , taking the cusp-limit  $\tau_0 \rightarrow i\infty$  in Theorem 3.2 collapses the  $S$ -segment onto the classical Mellin contour  $[0, i\infty]$  while via Corollary 3.4 the  $T$ -segment vanishes exponentially as  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(\tau_0) \sim e^{-2\pi\text{Im}(\tau_0)N}$  (as  $f \in S_k \Leftrightarrow f = \sum_{n \geq N} c_f(n)q^n$  with  $N \geq 1$ ). I.e., here, one recovers the classical L-integral of  $f$  without obstruction:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt. \quad (27)$$

This story changes when  $f = 1/q^N(1 + O(q)) \in S_k^!$  for some  $N \geq 1$ .

In this context, while of course the sum  $L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S) = L^*(f, s)$  is manifestly invariant under changes of the base-point, each of these two terms can indeed change values. Indeed, Corollary 3.4 clearly shows that as  $\tau_0 \rightarrow i\infty$ , when  $f \in S_k^!$  has negative powers of  $q$  in its  $q$ -series, then  $L_T^*(f, s; \gamma_{\tau_0}^T) \sim e^{2\pi N \text{Im}(\tau_0)}$ . Discarding  $\gamma_{\tau_0}^T$  therefore does violence to the result: When the  $S$ -contour is isolated from its  $T$ -partner, and  $\tau_0 \rightarrow i\infty$ , it diverges. This is the precise reason that [1] needed to regularize their divergent L-integral representation for properly weak cusp forms.

**Remark 3.6** (Recovering the classical L-integral for  $f \in M_k$ ). *In textbook treatments [4], when  $f \in M_k$ , its L-integral is defined as*

$$\begin{aligned} L^*(f, s)|_{\text{cl}} &:= \int_0^\infty (f(it) - c_f(0))t^{s-1}dt = \frac{\Gamma(s)}{(2\pi)^s} \sum_{n=1}^\infty \frac{c_f(n)}{n^s} \\ &= -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n=1}^\infty c_f(n) \left( \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) \end{aligned} \quad (28)$$

*Classically, this constant-term subtraction is needed because  $\int_0^\infty t^{s-1}dt$  is a pure divergence, so the  $q$ -series cannot commute with the integration contour.*

*However, as is clear from our general Theorem 3.2 for L-integrals for  $f \in M_k^!$ , and from the specific results of the  $S$ - and  $T$ -contours in Lemma 3.3 and Corollary 3.5, our L-integral requires no such explicit subtraction of the constant term  $-c_f(0)$ . Thus, not only does our prescription naturally handle divergences at poles when  $f \in M_k^!$  has polar terms in its  $q$ -series, but it also obviates the need to by-hand remove the constant mode from the kernel of the L-integral that is needed when defining L-functions in terms of the classical Mellin contour,  $\int_0^\infty$ .*

### 3.3 Period polynomials, periods and quasi-periods

Because  $L^*(f, s)$  is canonical for every  $f \in S_k^!$ , and because period polynomials for a given modular form can be written in terms of their L-functions evaluated at integer- $s$  in the critical strip, they need no separate construction.

**Definition 3.7** (Period polynomial for  $S_k^!$ ). *For  $f \in S_k^!$  and  $n = k - 2$ , the period polynomial is*

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (29)$$

**Proposition 3.8** (Period polynomial as a contour integral). *With  $(X - \tau Y)^n$  in place of the L-weight  $\tau^{s-1}$ , and  $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell+1) X^{n-\ell} Y^\ell$  in place of  $\tilde{k}_T$ , the same two segments assemble  $r_f$  directly:*

$$r_f(X, Y) = (2\pi i)^{n+1} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (30)$$

*Proof.* By Definition 2.3, the  $X^{n-\ell}Y^\ell$  coefficient of the bracket is

$$r_f|_{X^{n-\ell}Y^\ell} = (-1)^\ell \binom{n}{\ell} i^{\ell+1} [L_S^*(f, \ell+1) + L_T^*(f, \ell+1)] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (31)$$

This matches definition 3.7 term by term, and proves the claim.  $\square$

**Remark 3.9** (The role of the  $T$ -kernel in Eichler–Shimura theory). *The Hurwitz  $T$ -kernel in this note was originally uncovered in order to reproduce the manifestly finite computations of the quasi-periods for  $\hat{\Delta} = 1/q + O(q^2) \in S_{12}^1$  originally given in [2]. Its role in the period polynomial (30) is sharp: the  $S$ -segment alone is not invariant under the parabolic shift  $\tau_0 \mapsto \tau_0 + 1$ , and the  $T$ -segment  $\int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau$  is precisely the term that cancels this variation, leaving  $r_f(X, Y)$  independent of the basepoint  $\tau_0$  (Proposition 2.6). This basepoint-independence is the finite-contour counterpart of the parabolicity condition in Eichler–Shimura theory.*

Having now defined the period polynomial associated to a cusp form  $f \in S_k^!$ , we now seek to find its (quasi-)periods, which we define presently. First, though, we briefly mention some important facts from the theory of period polynomials, but refer the newcomer to the field to standard references such as [6] for more systematic discussions. To begin, because period polynomials are integrals of the  $\mathrm{SL}_2(\mathbb{Z})$  covariant differential,<sup>1</sup>

$$\omega_f := f(\tau)(X - \tau Y)^n d\tau. \quad (32)$$

Due to its origin as an integral of this covariant differential, it must satisfy two nontrivial relations

$$r_f(X, Y)|_{(1+S)} = r_f(X, Y)|_{(1+U+U^2)} = 0, \quad (33)$$

where  $U := TS$  and  $r_f(X, Y)|_\gamma := r_f(aX + bY, cX + dY)$  for all  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ , and  $r_f|_{\gamma+\sigma} := r_f|_\gamma + r_f|_\sigma$ . These two constraints together impose extremely stringent constraints on  $r_f(X, Y) \in \mathbb{C}[X, Y]_{k-2}$ . Specifically, they require it to lie within a  $2 \dim(S_k) + 1$ -dimensional subspace of the  $k - 2$ -dimensional space  $\mathbb{C}[X, Y]_{k-2}$ . When  $\dim(S_k) = 1$ , this 3-dimensional space is spanned by the three polynomials, whose definition we (hopefully correctly lifted) from [6, §1.3]:

**Definition 3.10.** For  $k$  with  $\dim S_k = 1$ ,  $n := k - 2$ ,  $p_0 := X^n - 1$ ,  $B_i = \oint t^{j-1} e^t / (t - 1) dt$ , and  $B_m^0(X) := \sum_{i \neq 1} \binom{m}{i} B_i X^{m-i}$ . The even and odd Kohnen–Zagier bases [6, Thm. 1'] are

$$W_-^{(k)}(X) := \mathrm{pp} \left( \frac{1}{n-1} (B_{n-1}^0(X) + X^n B_{n-1}^0(\frac{1}{X})) - \frac{1}{3} (B_3^0(X) + X^n B_3^0(\frac{1}{X})) \right), \quad (34)$$

$$W_+^{(k)}(X) := \mathrm{pp} \left( \frac{1}{2} (B_2^0(X) - X^n B_2^0(\frac{1}{X})) + \frac{1}{n} (B_n^0(X) - X^n B_n^0(\frac{1}{X})) - c p_0 \right), \quad (35)$$

where  $\mathrm{pp}$  act on any polynomial with rational coefficients by multiplying it by all coefficient-denominators and dividing by the  $\mathrm{gcd}$ , and  $c$  eliminates the  $p_0$  component of  $W_+^{(k)}$ .

**Definition 3.11** (Periods and Quasi-Periods). Let  $k$  be such that  $\dim(S_k) = 1$ ,  $n := k - 2$ , let  $W_\pm^{(k)}(X, Y) := Y^n W_\pm^{(k)}(X/Y)$  be as in Definition 3.10, and let  $r_f(X, Y)$  be as in Definition 3.7. Further, define the two numbers, projections

$$\omega^\pm(f) = \frac{\langle r_f, W_\pm^{(k)} \rangle}{\langle W_\pm^{(k)}, W_\pm^{(k)} \rangle}, \quad \langle P, Q \rangle = \sum_a (-1)^a \binom{n}{a}^{-1} p_a q_{n-a}. \quad (36)$$

When  $f$  is a cusp form,  $\omega^\pm(f)$  are its periods; when  $f$  is a weak cusp form,  $\omega^\pm(f)$  are its quasi-periods.

---

<sup>1</sup>Technically, period polynomials are the sum of contour integrals  $\int_{\gamma^{-1}(\tau_0)}^{\tau_0} \omega_f$ , plus an additional piece which fixes parabolicity. This extra piece is the focus of Remark 3.9.

Now, having defined our periods and quasi-periods, we use them together with our L-functions to extract periods and quasi-periods for  $f \in S_k^!$  (when  $\dim(S_k) = 1$ ). To this effect, we recall the notation of [5]. Decomposing the modular weight  $k = 12\lambda + k'$  where  $k' \in \{0, 4, 6, 8, 10, 14\}$ , we see  $\lambda = \dim(S_k)$  when  $\lambda \geq 0$ . For any  $\lambda$ , we may denote  $f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$ . When  $\dim(S_k) = 1$ , we have  $f_{k,m} = 1/q^m + O(q^2)$ . Here, define  $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$  and  $\hat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$ . Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\hat{\Delta}_k) & \eta^-(\hat{\Delta}_k) \end{pmatrix}, \quad \alpha_k := \frac{(-1)^{1+k/2} \det(\text{pd}_k)}{(2\pi i)^{k-1} (k-2)!} \quad (37)$$

Brown–Hain [7, Prop. 5.6] show that  $\det(\text{pd}_k)$  satisfies a Legendre-type relation,  $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$ . Brown [2, §8.3] numerically finds  $\alpha_k = 1$ ; we reproduce this result at  $k = 12$  and give the analogous rational multiples for  $k \in \{16, 18, 20, 22, 26\}$  in Table 1, below. The value of  $\alpha_k$  is tied to the normalization of Definition 3.10: our primitive-integer choice reproduces Brown’s  $\alpha_{12} = 1$ , whereas rescaling the same lines to the Kohnen–Zagier polynomials  $r^\pm(R_n)$  [6, Thm. 1’]—whose  $R_n$  normalization is itself  $n$ -dependent—gives different  $\alpha_k$  and spoils  $\alpha_{12} = 1$ . The  $\alpha_k$  are therefore basis-relative rationals.

| $k$ | $\omega^+(\Delta_k)$ | $\omega^-(\Delta_k)/i$ | $\eta^+(\hat{\Delta}_k)$ | $\eta^-(\hat{\Delta}_k)/i$ | $\alpha_k$ |
|-----|----------------------|------------------------|--------------------------|----------------------------|------------|
| 12  | −6.89168e7           | −5.58502e6             | 1.27202e11               | 1.02767e10                 | 1          |
| 16  | −8.81481e11          | −1.54063e10            | 4.14120e13               | 6.83541e11                 | 13/30      |
| 18  | −5.07833e13          | −5.43305e12            | −2.19200e14              | −3.36339e13                | 2/3        |
| 20  | −4.22882e16          | −3.73170e14            | 6.38640e16               | −1.89823e14                | 17/5       |
| 22  | −1.49974e18          | −4.24489e16            | 1.43795e18               | −4.41050e16                | 19/21      |
| 26  | −1.18673e22          | −3.60961e21            | 1.18653e22               | −3.61016e21                | 23/15      |

Table 1: Periods  $\omega^\pm(\Delta_k)$  and quasi-periods  $\eta^\pm(\hat{\Delta}_k)$ , and the rational number  $\alpha_k$ . While  $\alpha_{12} = 1$  matches Brown [2],  $\alpha_k$  for  $k \geq 16$  appear to be new, interesting, rational numbers, albeit with implicit dependence on the period-polynomial basis normalizations—in contrast to the normalization-free Petersson check above. We have not found a clear pattern for the  $\alpha_k$ . See [9] for the scripts that extracted these figures (to 50-digit precision).

## 4 Meromorphic modular forms

For  $f \in F_k$  with poles in  $\mathbb{H}$ , the two segment integrals of Definition 2.3 depend on their contours; we show their sum depends only on the homotopy class of the pair.

**Proposition 4.1.** *For  $f \in F_k$  and  $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , the L-integral  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  depends only on the homotopy class of its two segments in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , and not on the basepoint  $\tau_0$ . At fixed  $\tau_0$  each of  $L_S^*, L_T^*$  is separately path-independent; a change of basepoint moves value between the two, leaving only their sum invariant.*

*Proof.* At fixed  $\tau_0$  the integrands  $f(\tau)\tau^{s-1}$  and  $f(\tau)\tilde{k}_T(\tau, s)$  are holomorphic on  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , so Cauchy’s theorem makes each of  $L_S^*, L_T^*$  constant on the homotopy class of its own segment. Under a motion of the basepoint,  $\partial_{\tau_0} L^*$  is evaluated at the moving endpoints alone; the computation of Proposition 2.6 — which uses only  $f(\tau_0 - 1) = f(\tau_0)$  and  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$  — cancels the boundary terms of  $L_S^*$  against those of  $L_T^*$ , the interior poles lying off the contours. Hence  $\partial_{\tau_0} L^* = 0$  along any motion in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , while each summand separately varies.  $\square$

**Definition 4.2.** *For  $f \in F_k$  and  $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ , let  $[\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]$  denote the homotopy classes of the two segments in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ .*

Thus, with  $\gamma_{\tau_0}^S \in [\gamma_{\tau_0}^S]$  and  $\gamma_{\tau_0}^T \in [\gamma_{\tau_0}^T]$ , by Proposition 4.1 the value

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) := L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) \quad (38)$$

is independent of the chosen representative paths and independent of  $\tau_0$  (subject to natural constraints of  $\tau_0$  monodromies around singular points, which move between homotopy classes).

We would like to speak of “the” L-integral. Homotopy classes of course split this into L-integrals for contours in different homotopy classes. And, so, in what follows, we define the reference homotopy classes:

**Definition 4.3.** Let  $f \in F_k$ ,  $\tau_0$  be the base-point of  $L^*(f, s)$  in the usual way, and let  $f^{-1}(\infty)$  be the full set of all poles of  $f$ . Further, let  $\tau_p^*$  denote the pole of lowest height at/above the  $\text{Im}(\tau) = 1$  line, and denote its imaginary height by  $y^* := \text{Im}(\tau_p^*)$ . If  $y^* = 1$ , then let  $\tau_p^\dagger = x^\dagger + iy^\dagger$  denote  $f$ ’s pole of next lowest height above  $\tau_p^*$ . I.e.,  $y^* < y^\dagger$ ; if no such pole exists, then  $y^\dagger := \infty$ .

We define the minimal homotopy class of  $f$ , which we denote  $[\gamma_f] = [\gamma_f^T] + [\gamma_f^S]$ , as follows.

1. If  $y^* > 1$ , then  $[\gamma_f^T]$  is the homotopy class containing the horizontal line from  $i - 1$  to  $i$ , and  $[\gamma_f^S]$  is the homotopy class containing the trivial path from  $i$  to itself.
2. If  $y^* = 1$ , let  $\delta$  be any positive constant such that  $y^* < 1 + \delta < y^\dagger$ . Let  $\tau_0^* := i + i\delta$ . Here,  $[\gamma_f^T]$  is the homotopy class containing the horizontal line from  $\tau_0^* - 1$  to  $\tau_0^*$ , and  $[\gamma_f^S]$  is the homotopy class containing the almost-straight-line path that goes from  $-1/\tau_0^*$  to  $\tau_0^*$  along the imaginary- $\tau$  axis except when it passes a pole on the axis. When there is a pole on the imaginary- $\tau$  axis e.g. at  $\tau_p = i$ , this special contour  $\gamma_{\tau_0^*}^S$  passes to the left of such a pole.

With this definition, we may thus unambiguously define

$$L^*(f, s) := L^*(f, s; [\gamma_f^S], [\gamma_f^T]) . \quad (39)$$

Now, having fixed the reference value  $L_0^*(f, s)$  in the trivial homotopy class in Definition 4.3, now consider two paths  $\gamma_{\tau_0}^S$  and  $\gamma_{\tau_0}^T$  which may be in any homotopy class. Further, let  $X_S(p), X_T(p) \in \mathbb{Z}$  count the windings  $\gamma_{\tau_0}^S$ , resp.  $\gamma_{\tau_0}^T$  has about each pole  $\tau_p \in f^{-1}(\infty)$  relative to the null-classes.

**Lemma 4.4.** Let  $f \in F_k$ ,  $r_S(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tau^{s-1})$  and  $r_T(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tilde{k}_T(\tau, s))$ . Then,

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) = L_0^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_{\tau_p \in f^{-1}(\infty)} \left[ X_S(\tau_p) r_S(f, \tau_p) + X_T(\tau_p) r_T(f, \tau_p) \right]. \quad (40)$$

*Proof.* Two pairs whose winding data differ by one unit at a single pole  $p$ , on one segment, bound a loop encircling  $p$  once; the residue theorem adds  $2\pi i$  times the residue of that segment’s integrand —  $f\tau^{s-1}$  for  $\gamma^S$ ,  $f\tilde{k}_T$  for  $\gamma^T$  — and the prefactor  $e^{-\pi i s/2}$  of Definition 2.3 multiplies through. Within-class constancy is Proposition 4.1; summing over crossings gives (40).  $\square$

The polylog projection renders every pole with  $\text{Im} \tau_p \geq 1$  as an explicit tail, leaving  $(I - \hat{P})f$  with all poles strictly below the line  $\text{Im} \tau = 1$ . This is forced for poles with  $\text{Im} \tau_p > 1$ : lying above  $\gamma_f^T$ , they make the  $q$ -series of  $f$  diverge on the contour ( $|c_f(n)| \gtrsim n^{m-1} e^{2\pi n \text{Im} \tau_p}$  for a pole of order  $m$  [3]). A pole on the line itself sits just below the lifted  $\gamma_f^T$  of Definition 4.3; we project it too, so the regularized part stays clean and it appears as a tail rather than a  $\gamma^S$ -integral around the pole.

**Definition 4.5.** Let  $f \in F_k$  have, at a pole  $\tau_p$  of order  $m_p$ , the Laurent principal part  $\sum_{m=1}^{m_p} a_{-m}^{(p)} (\tau - \tau_p)^{-m}$ , where  $a_{-m}^{(p)}$  is the coefficient of  $(\tau - \tau_p)^{-m}$ . Since  $\text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$  has a pole of order  $m$  at  $\tau_p$ , there are unique  $r_{f, \tau_p}^*(m)$ ,  $1 \leq m \leq m_p$ , for which the polylog projection

$$\widehat{P}_{\tau_p} f(\tau) := \sum_{m=1}^{m_p} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)}) \quad (41)$$

has the same principal part as  $f$  at  $\tau_p$  (a triangular, hence invertible, linear system relating  $r_{f, \tau_p}^*(m)$  to  $a_{-1}^{(p)}, \dots, a_{-m}^{(p)}$ ). The projection  $\widehat{P} := \sum_{\text{Im} \tau_p \geq 1} \widehat{P}_{\tau_p}$  removes the principal part of every pole with  $\text{Im} \tau_p \geq 1$ .

**Definition 4.6.** With  $\widehat{P}$  as in Definition 4.5, let  $c_{Rf}(n)$  be the Fourier coefficients of the regularized form  $(I - \widehat{P})f$  in its expansion valid for large  $\text{Im} \tau$ :

$$(I - \widehat{P})f(\tau) = \sum_n c_{Rf}(n) q^n, \quad q = e^{2\pi i \tau}. \quad (42)$$

**Definition 4.7.**  $F(\infty)$  is the space of meromorphic  $g$  on  $\mathbb{H}$  with  $g(\tau + 1) = g(\tau)$ , finitely many poles in each compact subset of  $\{\text{Im} \tau > 0, |\text{Re} \tau| \leq \frac{1}{2}\}$ , and  $g(\tau) = O(e^{C \text{Im} \tau})$  as  $\text{Im} \tau \rightarrow \infty$  for some  $C$ .

Then  $F_k \subset F(\infty)$ , and  $(I - \widehat{P})f \in F(\infty)$ : the polylog tails are  $T$ -periodic but not modular, so the projection leaves  $F_k$ . After projection every pole of  $(I - \widehat{P})f$  lies strictly below  $\text{Im} \tau = 1$ ; pole heights accumulate only at 0, so there is a maximal one,  $y_b < 1$  (take  $y_b = 0$  if  $(I - \widehat{P})f$  is holomorphic).

**Lemma 4.8.** Let  $y_b < 1$  be the maximal pole height of  $(I - \widehat{P})f$ , and  $m_b$  the maximal order among poles at height  $y_b$ . There is a  $C > 0$  such that, for every  $\varepsilon > 0$ ,

$$C n^{m_b-1} e^{2\pi n y_b} \leq |c_{Rf}(n)| \leq e^{2\pi n(y_b + \varepsilon)}, \quad (43)$$

the upper bound for all large  $n$ , the lower for infinitely many  $n$ . In particular  $|c_{Rf}(n)| e^{-2\pi n} \rightarrow 0$  geometrically, at base  $e^{-2\pi(1-y_b-\varepsilon)}$ .

*Proof.*  $(I - \widehat{P})f$  is  $T$ -periodic and continuous on  $\text{Im} \tau = y$  for each  $y \in (y_b, 1]$ , so Parseval gives  $\int_{-1/2}^{1/2} |(I - \widehat{P})f(\sigma + iy)|^2 d\sigma = \sum_n |c_{Rf}(n) e^{-2\pi y n}|^2 < \infty$ , whence  $c_{Rf}(n) e^{-2\pi y n} \rightarrow 0$ ;  $y = y_b + \varepsilon$  gives the upper bound [3]. For the lower bound, subtract the polylog projection of the height- $y_b$  poles: the remainder's coefficients are  $o(e^{2\pi n y_b})$ , while the subtracted tails contribute  $\sum_j C_j n^{m_j-1} e^{-2\pi i n x_j} e^{2\pi n y_b}$  with  $C_j \neq 0$  at order  $m_b$ ; the  $x_j$  distinct mod 1, the mean value  $\sum_{m_j=m_b} |C_j|^2 > 0$  bounds the sum below along infinitely many  $n$ .  $\square$

For  $n \neq 0$ , the single-mode contribution of Theorem 3.2 is

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (44)$$

Each projected pole contributes a single polylog tail.

**Definition 4.9.** With  $\text{Li}_s(x) = \sum_{n \geq 1} x^n n^{-s}$ , the tail integral of a projected pole at  $z$  with  $\text{Im} z > 1$  is the  $L$ -integral of that tail in the minimal class of  $f$  (Definition 4.3):

$$I_N(s, z) := L^*(\text{Li}_{-N}(e^{2\pi i(\cdot - z)}), s; [\gamma_f^S], [\gamma_f^T]). \quad (45)$$

The tail's poles then lie above  $\gamma_f^S$ , whose contribution vanishes, so  $I_N(s, z)$  is the  $T$ -segment integral alone.

**Lemma 4.10.** For  $\text{Im } z > 1$ , polylog inversion  $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w)$  resums (45) into

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} G_{s,k}(-n), \quad (46)$$

convergent exponentially with base  $e^{-2\pi(\text{Im } z - 1)}$ .

*Proof.* By Proposition 4.1 we may evaluate  $I_N(s, z)$  in the  $\delta \rightarrow 0$  representative of the minimal class; there the tail — regular at  $i$  since  $\text{Im } z > 1$  — has vanishing  $\gamma_f^S$ -integral and  $\gamma_f^T \rightarrow [i-1, i]$ . On  $[i-1, i]$  one has  $|e^{2\pi i(\tau-z)}| > 1$ , so the defining series of  $\text{Li}_{-N}$  diverges; inversion rewrites it as  $(-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$ , convergent since  $|e^{2\pi i(z-\tau)}| < 1$ , and term-by-term integration against  $\tilde{k}_T$  gives  $\sum_n n^N e^{2\pi i n z} G_{s,k}(-n)$ , the mode- $(-n)$  block of Theorem 3.2; for  $N = 0$  the leftover  $-1$  in  $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$  contributes  $\frac{1}{s} + \frac{i^k}{k-s}$ . The rate is  $e^{-2\pi(\text{Im } z - 1)}$ , from  $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$  against  $|e^{2\pi i n z}| = e^{-2\pi n \text{Im } z}$ .  $\square$

For  $\text{Im } z > 1$ , write  $w := e^{2\pi i(z-i)}$ ,  $Q := e^{2\pi i z}$ , and  $(a)_{(j)} := a(a-1) \cdots (a-j+1)$ ,  $(a)_{(0)} := 1$  (principal branches).

**Definition 4.11.** Define  $\Pi_N(s; w)$ ,  $\Lambda_N(s; Q)$ ,  $\mathcal{E}_N(s, z)$  by

$$\Pi_N(s; w) := \sum_{j=0}^{N+1} \frac{(-1)^{j+1} [(s-1)_{(j)} + i^k(k-s-1)_{(j)}]}{(2\pi)^{j+1}} \text{Li}_{j+1-N}(w), \quad (47)$$

$$\Lambda_N(s; Q) := \frac{\Gamma(s)}{(-2\pi)^s} \text{Li}_{s-N}(Q) + i^k \frac{\Gamma(k-s)}{(-2\pi)^{k-s}} \text{Li}_{k-s-N}(Q), \quad (48)$$

$$\mathcal{E}_N(s, z) := - \sum_{n \geq 1} n^N Q^n \left[ (s-1)_{(N+2)} \frac{\Gamma(s-N-2) - \Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s} + i^k (k-s-1)_{(N+2)} \frac{\Gamma(k-s-N-2) - \Gamma(k-s-N-2, -2\pi n)}{(-2\pi n)^{k-s}} \right]. \quad (49)$$

Here  $\Pi_N$  is finite and carries the entire  $w \rightarrow 1$  singularity ( $\text{Li}_{\leq 1}(w)$ ); the  $\text{Li}_{s-N}(Q)$ ,  $\text{Li}_{k-s-N}(Q)$  in  $\Lambda_N$  are polylogarithms of non-integer order; and  $\mathcal{E}_N$  is absolutely convergent, its terms  $O(n^{-3}|w|^n)$ .

**Lemma 4.12.** With  $\Pi_N, \Lambda_N, \mathcal{E}_N$  as in Definition 4.11, for  $\text{Im } z > 1$  and every  $s \in \mathbb{C}$ ,

$$I_N(s, z) = \delta_{N,0} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} [\Pi_N(s; w) + \Lambda_N(s; Q) + \mathcal{E}_N(s, z)], \quad (50)$$

and the right side extends  $I_N(s, \cdot)$  continuously to the edge  $\text{Im } z = 1$  for  $z \notin i + \mathbb{Z}$ .

*Proof.* Insert into Lemma 4.10 the recursion  $\Gamma(s, x) = e^{-x} \sum_{j < J} (s-1)_{(j)} x^{s-1-j} + (s-1)_{(J)} \Gamma(s-J, x)$  at  $x = -2\pi n$ ,  $J = N+2$ , for both incomplete gammas in  $G_{s,k}(-n)$ . The  $e^{2\pi n}$  factors recombine with  $e^{2\pi i n z}$  into  $w^n$ , and  $\sum_n n^{N-j-1} w^n = \text{Li}_{j+1-N}(w)$  gives  $\Pi_N$ . Writing  $\Gamma(s-J, x) = \Gamma(s-J) - [\Gamma(s-J) - \Gamma(s-J, x)]$  and using  $(s-1)_{(J)} \Gamma(s-J) = \Gamma(s)$ , the complete parts resum to  $\Lambda_N$  via  $\sum_n n^{N-s} Q^n = \text{Li}_{s-N}(Q)$ , the lower-gamma parts to  $\mathcal{E}_N$ . From  $\Gamma(\sigma, x) \sim x^{\sigma-1} e^{-x}$ , the terms of  $\mathcal{E}_N$  are  $O(n^{N-J-1} e^{2\pi n}) |Q|^n = O(n^{-3} |w|^n)$ , uniform on  $\text{Im } z \geq 1$ ;  $\Lambda_N$  is continuous there since  $|Q| \leq e^{-2\pi}$ ; and each  $\text{Li}_{j+1-N}(w)$  is continuous up to  $|w| = 1$  away from  $w = 1$ .  $\square$

The one edge point the formula misses is  $z = i$ , where  $\Pi_N$  is singular ( $w = 1$ ); we compute  $I_N(s, i)$  directly. Its finiteness rests on a relation among the Laurent coefficients of  $f$  at the  $S$ -fixed point.

**Lemma 4.13.** Let  $f \in F_k$  have a pole of order  $P$  at  $i$ , with principal part  $\sum_{m=1}^P a_{-m}(\tau - i)^{-m}$ . Modularity  $f(-1/\tau) = \tau^k f(\tau)$  is equivalent to

$$(-1)^r \sum_{m=r}^P a_{-m} \binom{m}{r} i^{m-r} = i^k \sum_{m=r}^P a_{-m} \binom{k}{m-r} (-i)^{m-r}, \quad r = 1, \dots, P. \quad (51)$$

The leading relation ( $r = P$ ) is  $((-1)^P - i^k) a_{-P} = 0$ , so the pole order matches  $k \bmod 4$  ( $P$  odd  $\Leftrightarrow k \equiv 2$ ,  $P$  even  $\Leftrightarrow 4 \mid k$ ); at  $P = 1$  this is  $(1 + i^k) a_{-1} = 0$ , and for a double pole the  $r = 1$  relation is  $a_{-1} = \frac{i(k-2)}{2} a_{-2}$ .

*Proof.* Write  $\tau = i + \varepsilon$ . Then  $-1/\tau - i = -\varepsilon/(1 - i\varepsilon)$ , so the principal part of  $f(-1/\tau)$  is  $\sum_m a_{-m} (i - \varepsilon^{-1})^m$  and that of  $\tau^k f(\tau)$  is  $i^k (1 - i\varepsilon)^k \sum_m a_{-m} \varepsilon^{-m}$ . Matching the coefficient of  $\varepsilon^{-r}$  through  $(i - \varepsilon^{-1})^m = \sum_j \binom{m}{j} i^{m-j} (-1)^j \varepsilon^{-j}$  and  $(1 - i\varepsilon)^k = \sum_l \binom{k}{l} (-i)^l \varepsilon^l$  gives (51).  $\square$

**Definition 4.14.** For  $N \geq 0$ , with  $H_N := \sum_{r=1}^N \frac{1}{r}$  and  $c_j^+(s) := (s-1)_{(j)} + i^k (k-s-1)_{(j)}$ ,

$$\text{FP}_N(s, k) := \frac{c_N^+(s) H_N}{(2\pi)^{N+1}} + \sum_{\substack{j \geq 0 \\ j \neq N}} \frac{c_j^+(s) \zeta(j+1-N)}{(2\pi)^{j+1}} + \frac{(-1)^N N!}{(2\pi)^{N+1}} \sum_{t=0}^{N-1} \frac{(-1)^t (s-1)_{(t)}}{t! (N-t)} + \mathcal{E}_N^+(s), \quad (52)$$

the  $j$ -sum taken to any  $J \geq N+2$ , with  $\mathcal{E}_N^+(s) := \sum_{n \geq 1} n^N e^{2\pi n} \left[ (s-1)_{(J)} \frac{\Gamma(s-J, 2\pi n)}{(2\pi n)^s} + i^k (k-s-1)_{(J)} \frac{\Gamma(k-s-J, 2\pi n)}{(2\pi n)^{k-s}} \right]$ .

**Lemma 4.15.** Let  $f \in F_k$  have a pole of order  $P$  at  $i$  with principal part  $\sum_{m=1}^P a_{-m}(\tau - i)^{-m}$ , and  $\widehat{P}_i f$  its projection. The pole's contribution to  $L^*(f, s)$  in the minimal class is

$$L^*(\widehat{P}_i f, s) = -2\pi i a_{-1} \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{m=1}^P \frac{(-2\pi i)^m a_{-m}}{(m-1)!} \text{FP}_{m-1}(s, k), \quad (53)$$

entire in  $s$  but for the simple poles at  $s = 0, k$  carried by  $a_{-1}$ ; the individually divergent contour pieces cancel by the relations of Lemma 4.13.

*Proof.* Evaluate on the minimal-class contour with basepoint  $i(1 + \delta)$  and let  $\delta \rightarrow 0$ . On the  $T$ -segment the pole at  $i$  lies below the contour; inverting each  $\text{Li}_{1-m}(e^{2\pi i(\tau-i)})$  and integrating against  $\widetilde{k}_T$  yields the harmonic-,  $\zeta$ -, and  $\mathcal{E}^+$ -terms of  $\text{FP}_{m-1}$  together with a part  $\sim \delta^{-(m-1)}, \dots, \log \delta$ ; the  $S$ -segment, passing to the left of  $i$ , gives the mirror divergences. Summed over  $m$  with weights  $r^*(m) = (-2\pi i)^m a_{-m}/(m-1)!$ , the divergent parts form a linear combination of the  $a_{-m}$  that vanishes precisely by the modularity relations of Lemma 4.13, leaving the  $\text{FP}$ -sum of (53); the simple part ( $m = 1$ ) additionally carries the  $-1$ -leftover pole  $-2\pi i a_{-1}(\frac{1}{s} + \frac{i^k}{k-s})$ , as in Lemma 4.10. The cutoff  $J$  is immaterial: raising it moves terms between the  $\zeta$ -sum and  $\mathcal{E}_N^+$ , which remains convergent as in Lemma 4.12.  $\square$

**Theorem 4.16.** Let  $f \in F_k$  have poles  $\tau_p$  of order  $P_p$  with  $\text{Im } \tau_p \geq 1$  in the strip  $|\text{Re } \tau| \leq \frac{1}{2}$ , regularized Fourier coefficients  $c_{Rf}(n)$  (Definition 4.6), and projection weights  $r_{f, \tau_p}^*(m)$  (Definition 4.5). In the minimal class,

$$L^*(f, s) = -c_{Rf}(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\tau_p \neq i} \sum_{m=1}^{P_p} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p) + L^*(\widehat{P}_i f, s), \quad (54)$$

with  $I_{m-1}$  from Lemmas 4.10–4.12 and the last term (present only for a pole at  $i$ ) from Lemma 4.15. The right side is entire in  $s$  apart from at most simple poles at  $s = 0$  and  $s = k$ , and reduces to the weakly-holomorphic  $L^*$  of Theorem 3.2 when  $f$  is holomorphic on  $\mathbb{H}$ .

*Proof.* Split  $f = (I - \hat{P})f + \hat{P}f$ . By Lemma 4.8 the Fourier series of the regularized part converges on  $\gamma_f^T$ ; Theorem 3.2 applied term by term gives its  $L^*$  as the first two sums. The projection  $\hat{P}f = \sum_{\tau_p} \sum_m r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$  contributes, for each  $\tau_p \neq i$ , the tail integrals  $I_{m-1}(s, \tau_p)$  of Definition 4.9, evaluated by Lemmas 4.10–4.12; the pole at  $i$  contributes  $L^*(\hat{P}_i f, s)$  of Lemma 4.15. Entirety and the pole locations follow from those of  $G_{s,k}$  and the  $\delta_{N,0}$  terms; when  $f$  is holomorphic on  $\mathbb{H}$ ,  $\hat{P}f = 0$  and  $c_{Rf} = c_f$ , recovering Theorem 3.2.  $\square$

## A Explicit period-polynomial bases

In this Appendix, we provide the explicit information needed to extract the quasi-periods presented in the main body of the note. For each weight  $k$  with  $\dim S_k = 1$ ,  $W_{\pm}^{(k)}$  are integer-coefficient generators of the even and odd cuspidal period polynomials, in the sense of Kohnen–Zagier [6, §1.3, Thm. 1']. The two sides are not symmetric. The odd  $W_{-}^{(k)}$  is a rational multiple of the Kohnen–Zagier odd period polynomial directly. The even  $W_{+}^{(k)}$  is the *cuspidal part* of the even one: their even period polynomial carries an Eisenstein term proportional to  $p_0 = X^{k-2} - Y^{k-2}$ , and  $W_{+}^{(k)}$  is that polynomial with the  $p_0$  term removed. Alongside are the period polynomials of the cusp form  $\Delta_k = f_{k,-1} = q + O(q^2)$  and the weak cusp form  $\hat{\Delta}_k = f_{k,1} = q^{-1} + O(q^2)$ , obtained from the finite-contour  $L$ -integral (Definition 2.3) and projected onto  $W_{\pm}$  by the Haberland pairing  $\langle P, Q \rangle = \sum_a (-1)^a \binom{n}{a}^{-1} p_a q_{n-a}$ ,  $n = k - 2$ . The periods  $\omega^{\pm}(\Delta_k)$  and quasi-periods  $\eta^{\pm}(\hat{\Delta}_k)$  are quoted to ten digits (full precision in [9], `period.polynomial.bases.py`).

**Comment on the sizes of the  $L$ -values.** One might expect the weak cusp form  $\hat{\Delta}_k = q^{-1} + O(q^2)$  to have far larger special values than the cusp form  $\Delta_k = q + O(q^2)$ : its  $q$ -series begins at  $n = -1$ , where the incomplete Gamma  $\Gamma(s, 2\pi n) \sim (2\pi n)^{s-1} e^{-2\pi n}$  of Theorem 3.2 is a factor  $\sim e^{4\pi}$  larger than the cusp form's  $n = 1$  term. Table 2 shows this fails on two counts. First, for a weak cusp form the leading term does not set the scale of the  $L$ -value: the rapidly growing coefficients  $c_{\hat{\Delta}_k}(n) \sim e^{4\pi\sqrt{n}}$  spread it across many modes. Second, the size of each mode is set not by the  $q$ -term  $\Gamma(s, 2\pi n)$  but by its companion  $\Gamma(k-s, 2\pi n)/(2\pi n)^{k-s} \sim (k-s-1)!/(2\pi)^{k-s}$ , whose factorial growth in the weight is common to both forms and dominates as  $k$  grows, driving the two special values to the same magnitude. Indeed, the ratio of the  $L$ -values of  $\Delta_k$  and  $\hat{\Delta}_k$  become very similar in magnitude as for larger  $k$ .

| $f$                 | $c_f(-1)G(-1)$ | $c_f(1)G(1)$ | $c_f(2)G(2)$             | $c_f(3)G(3)$             | $c_f(4)G(4)$             | $c_f(5)G(5)$             | $\sum_{\text{disp}}$ | $L^*(f, 1)$ |
|---------------------|----------------|--------------|--------------------------|--------------------------|--------------------------|--------------------------|----------------------|-------------|
| $\Delta_{12}$       | 0              | 0.0059859    | $-2.7182 \times 10^{-5}$ | $2.5668 \times 10^{-7}$  | $-1.8534 \times 10^{-9}$ | $8.5164 \times 10^{-12}$ | 0.005959             | 0.005959    |
| $\hat{\Delta}_{12}$ | -116.87        | 0            | 54.035                   | 40.603                   | 9.5097                   | 1.074                    | -11.647              | -11.57      |
| $\Delta_{16}$       | 0              | 0.092931     | 0.00049966               | $-4.5711 \times 10^{-6}$ | $2.0693 \times 10^{-8}$  | $1.0307 \times 10^{-10}$ | 0.093426             | 0.093426    |
| $\hat{\Delta}_{16}$ | -110.83        | 0            | 27.755                   | 48.849                   | 23.929                   | 4.8955                   | -5.3965              | -4.808      |
| $\Delta_{18}$       | 0              | -0.5639      | 0.0018202                | $4.2075 \times 10^{-6}$  | $-1.0197 \times 10^{-7}$ | $6.9075 \times 10^{-10}$ | -0.56208             | -0.56208    |
| $\hat{\Delta}_{18}$ | -61.399        | 0            | 15.967                   | 26.4                     | 13.116                   | 2.7729                   | -3.1419              | -2.7957     |
| $\Delta_{20}$       | 0              | 4.3746       | 0.0037257                | 0.00010963               | $-5.9575 \times 10^{-7}$ | $-5.4512 \times 10^{-9}$ | 4.3784               | 4.3784      |
| $\hat{\Delta}_{20}$ | -110.99        | 0            | 12.388                   | 37.794                   | 37.073                   | 13.897                   | -9.834               | -6.9431     |
| $\Delta_{22}$       | 0              | -42.106      | 0.0055972                | 0.00029786               | $2.4712 \times 10^{-6}$  | $-2.3041 \times 10^{-8}$ | -42.1                | -42.1       |
| $\hat{\Delta}_{22}$ | -23.329        | 0            | 8.2814                   | 22.828                   | 22.03                    | 8.4356                   | 38.247               | 40.066      |
| $\Delta_{26}$       | 0              | -6889.8      | 0.0098301                | 0.001365                 | $7.8801 \times 10^{-5}$  | $1.2671 \times 10^{-6}$  | -6889.8              | -6889.8     |
| $\hat{\Delta}_{26}$ | 6821.8         | 0            | 4.4956                   | 14.892                   | 24.42                    | 16.468                   | 6882.1               | 6888.4      |

Table 2: Mode-by-mode decomposition of  $L^*(f, 1)$  for the cusp form  $\Delta_k$  and weak cusp form  $\hat{\Delta}_k$  at each weight with  $\dim S_k = 1$ . Each entry is the contribution  $c_f(n)G_{1,k}(n)$  of the  $q^n$  mode (Theorem 3.2, Eq. (44)), and  $\sum_{\text{disp}}$  sums the six displayed modes. The cusp form is reproduced by its  $n = 1$  mode alone; the weak form's rapidly growing coefficients spread the value across many modes until  $n = -1$  dominates at large  $k$ . The two  $L^*(f, 1)$  in each weight-pair approach equal magnitude (and opposite sign) as  $k$  grows.

**The  $c_T$  term in the even weak-form entry.** The weak form's period polynomial  $r_{\hat{\Delta}_k}$  is, like the cusp form's, annihilated by  $1 + S$  and  $1 + U + U^2$ , so it too lies in the period space  $\langle W_+, p_0, W_- \rangle$ ; but it carries a component along the Eisenstein polynomial  $p_0 = X^{k-2} - Y^{k-2}$  that the cusp form lacks. Since  $p_0$  occupies only the two extreme monomials  $X^{k-2}, Y^{k-2}$ —the  $s = 1$  and  $s = k - 1$  coefficients—and  $W_+$  is not orthogonal to  $p_0$  under the Haberland pairing, the projection  $\eta^+$  of (36) absorbs part of this Eisenstein content but not all. The unabsorbed remainder surfaces as an additive constant  $c_T$  in the extreme even coefficient of  $r_{\hat{\Delta}_k}$  in the matrices below. It occurs only there: the odd sector has no Eisenstein direction (so  $\eta^-$  is exact), the cusp form has no  $p_0$ -component (so no  $c_T$ ), and, lying outside the projection,  $c_T$  enters neither  $\eta^\pm$  nor  $\alpha_k$ .

#### Weight 12:

$$W_+^{(12)}(X, Y) = 36(X^{10} - Y^{10}) - 691(X^8 Y^2 - X^2 Y^8) + 2073(X^6 Y^4 - X^4 Y^6), \quad (55)$$

$$W_-^{(12)}(X, Y) = 4(X^9 Y + X Y^9) - 25(X^7 Y^3 + X^3 Y^7) + 42X^5 Y^5, \quad (56)$$

$$r_{\Delta_{12}} = 0.0001655268053 W_+^{(12)} + 0.009269276162 W_-^{(12)}, \quad (57)$$

$$r_{\hat{\Delta}_{12}} = -0.3061537142 W_+^{(12)} - 17.05597275 W_-^{(12)}. \quad (58)$$

These coefficients, localizing  $r_{\Delta_{12}}$  and  $r_{\hat{\Delta}_{12}}$  in the  $(W_+, W_-)$ -plane, translate to the periods and quasi-periods of Table 1 as follows:

$$\begin{pmatrix} [X^{10}] r_{\hat{\Delta}_{12}} & [X^9 Y] r_{\hat{\Delta}_{12}} \\ [X^{10}] r_{\Delta_{12}} & [X^9 Y] r_{\Delta_{12}} \end{pmatrix} = \begin{pmatrix} L^*(\hat{\Delta}_{12}, 1) & 10 L^*(\hat{\Delta}_{12}, 2) \\ L^*(\Delta_{12}, 1) & 10 L^*(\Delta_{12}, 2) \end{pmatrix} \quad (59)$$

$$= \begin{pmatrix} -11.5697 & -68.2239 \\ 5.95897 \times 10^{-3} & 3.70771 \times 10^{-2} \end{pmatrix} = \begin{pmatrix} 36 \eta^+ + c_T & 4 \eta^- \\ 36 \omega^+ & 4 \omega^- \end{pmatrix},$$

$$\begin{pmatrix} \eta^+ & \eta^- \\ \omega^+ & \omega^- \end{pmatrix} = \begin{pmatrix} -0.3061537 & -17.05597 \\ 1.655268 \times 10^{-4} & 9.269276 \times 10^{-3} \end{pmatrix}. \quad (60)$$

The second matrix collects the periods and quasi-periods themselves — the coefficients of  $r_{\Delta_{12}}, r_{\widehat{\Delta}_{12}}$  in the  $W_{\pm}^{(12)}$  basis, obtained by the Haberland projection and hence coboundary-free. Comparing the two: the odd entries agree ( $4\eta^-, 4\omega^-$  exactly), but the even weak-form entry of the raw matrix exceeds  $36\eta^+$  by  $c_T = -0.548$  — the  $C_T$ -coboundary, which contaminates every odd- $s$  value of  $\widehat{\Delta}_{12}$  and is removed only by the projection. Notably, at this weight the ratio of special  $L$ -values at  $s = 1$  is  $L^*(\widehat{\Delta}_{12}, 1)/L^*(\Delta_{12}, 1) \approx -1942$ , in accord with Table 2.

### Weight 16:

$$W_+^{(16)}(X, Y) = 360(X^{14} - Y^{14}) - 7234(X^{12}Y^2 - X^2Y^{12}) + 25319(X^{10}Y^4 - X^4Y^{10}) - 39787(X^8Y^6 - X^6Y^8), \quad (61)$$

$$W_-^{(16)}(X, Y) = 36(X^{13}Y + XY^{13}) - 245(X^{11}Y^3 + X^3Y^{11}) + 539(X^9Y^5 + X^5Y^9) - 660X^7Y^7, \quad (62)$$

$$r_{\Delta_{16}} = 0.0002595173464 W_+^{(16)} + 0.01640585888 W_-^{(16)}, \quad (63)$$

$$r_{\widehat{\Delta}_{16}} = -0.01227268476 W_+^{(16)} - 0.7278917244 W_-^{(16)}. \quad (64)$$

These coefficients, localizing  $r_{\Delta_{16}}$  and  $r_{\widehat{\Delta}_{16}}$  in the  $(W_+, W_-)$ -plane, translate to the periods and quasi-periods of Table 1 as follows:

$$\begin{pmatrix} [X^{14}] r_{\widehat{\Delta}_{16}} & [X^{13}Y] r_{\widehat{\Delta}_{16}} \\ [X^{14}] r_{\Delta_{16}} & [X^{13}Y] r_{\Delta_{16}} \end{pmatrix} = \begin{pmatrix} L^*(\widehat{\Delta}_{16}, 1) & 14 L^*(\widehat{\Delta}_{16}, 2) \\ L^*(\Delta_{16}, 1) & 14 L^*(\Delta_{16}, 2) \end{pmatrix} \quad (65)$$

$$= \begin{pmatrix} -4.80804 & -26.2041 \\ 9.34262 \times 10^{-2} & 5.90611 \times 10^{-1} \end{pmatrix} = \begin{pmatrix} 360\eta^+ + c_T & 36\eta^- \\ 360\omega^+ & 36\omega^- \end{pmatrix},$$

$$\begin{pmatrix} \eta^+ & \eta^- \\ \omega^+ & \omega^- \end{pmatrix} = \begin{pmatrix} -1.227268 \times 10^{-2} & -0.7278917 \\ 2.595173 \times 10^{-4} & 1.640586 \times 10^{-2} \end{pmatrix}. \quad (66)$$

with  $c_T = -0.390$  the  $C_T$ -coboundary in the even weak-form entry. Notably, at this weight the ratio of special  $L$ -values at  $s = 1$  is  $L^*(\widehat{\Delta}_{16}, 1)/L^*(\Delta_{16}, 1) \approx -51.5$ , in accord with Table 2.

### Weight 18:

$$W_+^{(18)}(X, Y) = 18000(X^{16} - Y^{16}) - 350936(X^{14}Y^2 - X^2Y^{14}) + 1096675(X^{12}Y^4 - X^4Y^{12}) - 1140542(X^{10}Y^6 - X^6Y^{10}), \quad (67)$$

$$W_-^{(18)}(X, Y) = 24(X^{15}Y + XY^{15}) - 154(X^{13}Y^3 + X^3Y^{13}) + 273(X^{11}Y^5 + X^5Y^{11}) - 143(X^9Y^7 + X^7Y^9), \quad (68)$$

$$r_{\Delta_{18}} = -3.122662695 \times 10^{-5} W_+^{(18)} - 0.1465501421 W_-^{(18)}, \quad (69)$$

$$r_{\widehat{\Delta}_{18}} = -0.0001378466906 W_+^{(18)} - 0.907233517 W_-^{(18)}. \quad (70)$$

These coefficients, localizing  $r_{\Delta_{18}}$  and  $r_{\widehat{\Delta}_{18}}$  in the  $(W_+, W_-)$ -plane, translate to the periods and quasi-periods of Table 1 as follows:

$$\begin{pmatrix} [X^{16}]r_{\widehat{\Delta}_{18}} & [X^{15}Y]r_{\widehat{\Delta}_{18}} \\ [X^{16}]r_{\Delta_{18}} & [X^{15}Y]r_{\Delta_{18}} \end{pmatrix} = \begin{pmatrix} L^*(\widehat{\Delta}_{18}, 1) & 16 L^*(\widehat{\Delta}_{18}, 2) \\ L^*(\Delta_{18}, 1) & 16 L^*(\Delta_{18}, 2) \end{pmatrix} \quad (71)$$

$$= \begin{pmatrix} -2.79574 & -21.7736 \\ -5.62079 \times 10^{-1} & -3.51720 \end{pmatrix} = \begin{pmatrix} 18000 \eta^+ + c_T & 24 \eta^- \\ 18000 \omega^+ & 24 \omega^- \end{pmatrix},$$

$$\begin{pmatrix} \eta^+ & \eta^- \\ \omega^+ & \omega^- \end{pmatrix} = \begin{pmatrix} -1.378467 \times 10^{-4} & -0.9072335 \\ -3.122663 \times 10^{-5} & -0.1465501 \end{pmatrix}. \quad (72)$$

with  $c_T = -0.315$  the  $C_T$ -coboundary in the even weak-form entry. Notably, at this weight the ratio of special  $L$ -values at  $s = 1$  is  $L^*(\widehat{\Delta}_{18}, 1)/L^*(\Delta_{18}, 1) \approx 4.97$ , in accord with Table 2.

### Weight 20:

$$\begin{aligned} W_+^{(20)}(X, Y) &= 26460(X^{18} - Y^{18}) - 523833(X^{16}Y^2 - X^2Y^{16}) \\ &\quad + 1746110(X^{14}Y^4 - X^4Y^{14}) - 2444554(X^{12}Y^6 - X^6Y^{12}) \\ &\quad + 2269943(X^{10}Y^8 - X^8Y^{10}), \end{aligned} \quad (73)$$

$$\begin{aligned} W_-^{(20)}(X, Y) &= 108(X^{17}Y + XY^{17}) - 715(X^{15}Y^3 + X^3Y^{15}) + 1452(X^{13}Y^5 + X^5Y^{13}) \\ &\quad - 1560(X^{11}Y^7 + X^7Y^{11}) + 1430X^9Y^9, \end{aligned} \quad (74)$$

$$r_{\Delta_{20}} = 0.0001654739435 W_+^{(20)} + 0.2549699125 W_-^{(20)}, \quad (75)$$

$$r_{\widehat{\Delta}_{20}} = -0.0002521273082 W_+^{(20)} + 0.1296973583 W_-^{(20)}. \quad (76)$$

These coefficients, localizing  $r_{\Delta_{20}}$  and  $r_{\widehat{\Delta}_{20}}$  in the  $(W_+, W_-)$ -plane, translate to the periods and quasi-periods of Table 1 as follows:

$$\begin{pmatrix} [X^{18}]r_{\widehat{\Delta}_{20}} & [X^{17}Y]r_{\widehat{\Delta}_{20}} \\ [X^{18}]r_{\Delta_{20}} & [X^{17}Y]r_{\Delta_{20}} \end{pmatrix} = \begin{pmatrix} L^*(\widehat{\Delta}_{20}, 1) & 18 L^*(\widehat{\Delta}_{20}, 2) \\ L^*(\Delta_{20}, 1) & 18 L^*(\Delta_{20}, 2) \end{pmatrix} \quad (77)$$

$$= \begin{pmatrix} -6.94306 & 14.0073 \\ 4.37844 & 27.5368 \end{pmatrix} = \begin{pmatrix} 26460 \eta^+ + c_T & 108 \eta^- \\ 26460 \omega^+ & 108 \omega^- \end{pmatrix},$$

$$\begin{pmatrix} \eta^+ & \eta^- \\ \omega^+ & \omega^- \end{pmatrix} = \begin{pmatrix} -2.521273 \times 10^{-4} & 0.1296974 \\ 1.654739 \times 10^{-4} & 0.2549699 \end{pmatrix}. \quad (78)$$

with  $c_T = -0.272$  the  $C_T$ -coboundary in the even weak-form entry. Notably, at this weight the ratio of special  $L$ -values at  $s = 1$  is  $L^*(\widehat{\Delta}_{20}, 1)/L^*(\Delta_{20}, 1) \approx -1.59$ , in accord with Table 2.

**Weight 22:**

$$W_+^{(22)}(X, Y) = 126000(X^{20} - Y^{20}) - 2485856(X^{18}Y^2 - X^2Y^{18}) \quad (79)$$

$$+ 8156715(X^{16}Y^4 - X^4Y^{16}) - 10564888(X^{14}Y^6 - X^6Y^{14})$$

$$+ 6603055(X^{12}Y^8 - X^8Y^{12}),$$

$$W_-^{(22)}(X, Y) = 360(X^{19}Y + XY^{19}) - 2366(X^{17}Y^3 + X^3Y^{17}) + 4641(X^{15}Y^5 + X^5Y^{15}) \quad (80)$$

$$- 4182(X^{13}Y^7 + X^7Y^{13}) + 1547(X^{11}Y^9 + X^9Y^{11}),$$

$$r_{\Delta_{22}} = -0.0003341270695 W_+^{(22)} - 0.7346639014 W_-^{(22)}, \quad (81)$$

$$r_{\widehat{\Delta}_{22}} = 0.0003197845612 W_+^{(22)} - 0.7633262888 W_-^{(22)}. \quad (82)$$

These coefficients, localizing  $r_{\Delta_{22}}$  and  $r_{\widehat{\Delta}_{22}}$  in the  $(W_+, W_-)$ -plane, translate to the periods and quasi-periods of Table 1 as follows:

$$\begin{pmatrix} [X^{20}] r_{\widehat{\Delta}_{22}} & [X^{19}Y] r_{\widehat{\Delta}_{22}} \\ [X^{20}] r_{\Delta_{22}} & [X^{19}Y] r_{\Delta_{22}} \end{pmatrix} = \begin{pmatrix} L^*(\widehat{\Delta}_{22}, 1) & 20 L^*(\widehat{\Delta}_{22}, 2) \\ L^*(\Delta_{22}, 1) & 20 L^*(\Delta_{22}, 2) \end{pmatrix} \quad (83)$$

$$= \begin{pmatrix} 40.0663 & -274.798 \\ -42.1000 & -264.479 \end{pmatrix} = \begin{pmatrix} 126000 \eta^+ + c_T & 360 \eta^- \\ 126000 \omega^+ & 360 \omega^- \end{pmatrix},$$

$$\begin{pmatrix} \eta^+ & \eta^- \\ \omega^+ & \omega^- \end{pmatrix} = \begin{pmatrix} 3.197846 \times 10^{-4} & -0.7633263 \\ -3.341271 \times 10^{-4} & -0.7346639 \end{pmatrix}. \quad (84)$$

with  $c_T = -0.227$  the  $C_T$ -coboundary in the even weak-form entry. Notably, at this weight the ratio of special  $L$ -values at  $s = 1$  is  $L^*(\widehat{\Delta}_{22}, 1)/L^*(\Delta_{22}, 1) \approx -0.952$ , in accord with Table 2.

**Weight 26:**

$$W_+^{(26)}(X, Y) = 34398000(X^{24} - Y^{24}) - 678984792(X^{22}Y^2 - X^2Y^{22}) \quad (85)$$

$$+ 2233675745(X^{20}Y^4 - X^4Y^{20}) - 2938319846(X^{18}Y^6 - X^6Y^{18})$$

$$+ 2062613685(X^{16}Y^8 - X^8Y^{16}) - 850046852(X^{14}Y^{10} - X^{10}Y^{14}),$$

$$W_-^{(26)}(X, Y) = 1080(X^{23}Y + XY^{23}) - 7106(X^{21}Y^3 + X^3Y^{21}) + 14025(X^{19}Y^5 + X^5Y^{19}) \quad (86)$$

$$- 13167(X^{17}Y^7 + X^7Y^{17}) + 7106(X^{15}Y^9 + X^9Y^{15}) - 1938(X^{13}Y^{11} + X^{11}Y^{13}),$$

$$r_{\Delta_{26}} = -0.0002002972716 W_+^{(26)} - 40.08330039 W_-^{(26)}, \quad (87)$$

$$r_{\widehat{\Delta}_{26}} = 0.0002002611462 W_+^{(26)} - 40.0894082 W_-^{(26)}. \quad (88)$$

These coefficients, localizing  $r_{\Delta_{26}}$  and  $r_{\widehat{\Delta}_{26}}$  in the  $(W_+, W_-)$ -plane, translate to the periods and quasi-periods of Table 1 as follows:

$$\begin{pmatrix} [X^{24}] r_{\widehat{\Delta}_{26}} & [X^{23}Y] r_{\widehat{\Delta}_{26}} \\ [X^{24}] r_{\Delta_{26}} & [X^{23}Y] r_{\Delta_{26}} \end{pmatrix} = \begin{pmatrix} L^*(\widehat{\Delta}_{26}, 1) & 24 L^*(\widehat{\Delta}_{26}, 2) \\ L^*(\Delta_{26}, 1) & 24 L^*(\Delta_{26}, 2) \end{pmatrix} \quad (89)$$

$$= \begin{pmatrix} 6888.42 & -43296.6 \\ -6889.83 & -43290.0 \end{pmatrix} = \begin{pmatrix} 34398000 \eta^+ + c_T & 1080 \eta^- \\ 34398000 \omega^+ & 1080 \omega^- \end{pmatrix},$$

$$\begin{pmatrix} \eta^+ & \eta^- \\ \omega^+ & \omega^- \end{pmatrix} = \begin{pmatrix} 2.002611 \times 10^{-4} & -40.08941 \\ -2.002973 \times 10^{-4} & -40.08330 \end{pmatrix}. \quad (90)$$

with  $c_T = -0.164$  the  $C_T$ -coboundary in the even weak-form entry. Notably, at this weight the ratio of special  $L$ -values at  $s = 1$  is  $L^*(\widehat{\Delta}_{26}, 1)/L^*(\Delta_{26}, 1) \approx -1.000$ , in accord with Table 2.

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